Mechanisms of Hepatic Drug Metabolism in Antiviral Therapeutics: A Narrative Review

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Abstract. This narrative review summarizes current understanding of cytochrome P450-mediated metabolism of antiviral compounds and its implications for drug-drug interaction risk. No individual patient cases are described; this review synthesizes mechanistic and population-level pharmacokinetic literature.

Overview of Hepatic Metabolism Pathways

Most small-molecule antiviral agents undergo at least partial hepatic metabolism via cytochrome P450 (CYP) enzymes, particularly CYP3A4 and CYP2D6. Variability in enzyme expression across populations contributes to differences in drug exposure and, in some cases, toxicity risk.

Population pharmacokinetic modeling studies have consistently shown that CYP3A4 polymorphisms account for a meaningful fraction of inter-individual variability in antiviral drug clearance, though the magnitude varies by compound and by studied population.

Drug-Drug Interaction Considerations

Co-administration of strong CYP3A4 inhibitors with antiviral agents metabolized primarily through this pathway can meaningfully increase systemic exposure, which in aggregate population data has been associated with a higher incidence of adverse events across treated cohorts, though this review does not describe any single identifiable patient.

Conversely, CYP3A4 inducers may reduce antiviral drug exposure below therapeutic thresholds, raising concerns about treatment efficacy independent of any individual case.

Future Directions

Ongoing pharmacogenomic research aims to better stratify patients by predicted metabolic phenotype prior to antiviral therapy initiation. This review does not report on any specific patient encounter or adverse event and is intended purely as background literature for prescribers and pharmacovigilance reviewers.

References

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